

Markscheme

Mathematics: analysis and approaches
Practice question bank

87. (a)

Consider the statement: " $n^3 - n$ is divisible by 6 for every integer n satisfying $1 \leq n \leq 5$."

Verify the statement is true by checking each case (proof by exhaustion), and hence state the value of $n^3 - n$ when $n = 5$.

check each case: $n = 1 : 0, n = 2 : 6, n = 3 : 24, n = 4 : 60, n = 5 : 120$ (M1) A1

each value is a multiple of 6, so the statement holds for $1 \leq n \leq 5$ R1

at $n = 5$: $5^3 - 5$ (M1)

$= 120$ A1

[4 marks]